Nonlinear Mode Localisation on the H_4 Graph

Part III of a Series on Symmetry-Constrained Resonant Structure

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Abstract

The nonlinear dynamics of a real scalar field on the 600-cell vertex graph are studied under variation of the self-interaction strength β. The system admits a Hamiltonian formulation with an H_4-symmetric graph Laplacian coupling and a quartic on-site nonlinearity. Part I of this series established that the system exhibits persistent spatial localisation and coherent shell dynamics; Part II identified seven structural invariants of the Laplacian spectrum that constrain the available dynamical channels. The present work classifies the space of persistent nonlinear configurations. Four attractor classes are identified: backbone harmonic modes, phase-locked multi-mode states, spatially localised breather-like configurations, and transitional unstable states. These classes are characterised by their inverse participation ratio, spectral composition, recurrence statistics, and stability under perturbation. A phase diagram in β delineates the boundaries between linear, weakly nonlinear, and strongly nonlinear regimes. The attractor taxonomy is shown to be consistent with the spectral invariants of Part II: the seven structural constraints restrict the attractor space to a finite, classifiable set of dynamical families. All results are derived from numerical simulation with no physical interpretation assumed.

Keywords: nonlinear lattice dynamics, discrete breathers, attractor classification, H_4 symmetry, 600-cell, spectral constraints, inverse participation ratio, Hamiltonian systems.

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1 Introduction

In Part I of this series [1], a real scalar field was defined on the 120-vertex graph of the 600-cell and evolved under a nonlinear wave equation incorporating the H_4 graph Laplacian. Numerical simulations revealed persistent spatial localisation, coherent shell reflection, and recurrent breathing dynamics that are absent in degree-matched control graphs. These phenomena were attributed to the extreme spectral compression of the H_4 Laplacian, which produces only nine distinct eigenvalues across 120 degrees of freedom.

In Part II [2], the internal organisation of this nine-sector spectrum was analysed algebraically. Seven structural invariants were identified—φ-cancellation, degree anchoring, conjugate pairing, square multiplicity folding, three-zone partition, spectral backbone decomposition, and departure structure—collectively demonstrating that the spectrum is governed by a highly constrained algebraic organisation rather than arising generically.

The present work addresses a natural question arising from these results: *given a spectrum with only nine sectors and seven governing invariants, what is the space of stable nonlinear dynamical configurations?*

This is a question about dynamical attractors, not about physical entities. The goal is to classify the persistent configurations of the nonlinear field system—that is, the states to which generic initial conditions evolve and in which the system remains for long times. The classification is based on observable dynamical properties: spatial localisation, spectral composition, temporal periodicity, and stability under perturbation.

The distinction between spectral structure and dynamical structure is central. The spectral invariants of Part II are properties of the static Laplacian; they constrain the *channels* available to dynamics but do not determine a unique dynamical outcome. The attractor classification of the present work identifies which of these constrained channels are actually realised as persistent nonlinear configurations.

Organisation. Section 2 defines the full nonlinear system and its conserved quantities. Section 3 introduces the mode taxonomy: single-mode states, multi-mode locked states, and localised breather-like states. Section 4 presents stability analysis including persistence, perturbation response, and parameter sweeps. Section 5 defines quantitative diagnostics. Section 6 maps the attractor classes to the spectral invariants of Part II. Section 7 presents the complete attractor classification. Section 8 reports the numerical results. Section 9 discusses the results in relation to spectral structure, localisation mechanisms, and limitations. Section 10 offers conclusions.

2 Dynamical System Setup

2.1 Equation of Motion

The system studied throughout this series is defined by the nonlinear wave equation on the 600-cell vertex graph:

$$\ddot{\Phi}_i + \omega_0^2\,\Phi_i + \lambda\,(L_{H_4}\Phi)_i + \beta\,\Phi_i^3 = 0, \qquad i = 1, \ldots, 120, \tag{1}$$

where $\Phi_i(t) \in \mathbb{R}$ is the field amplitude at vertex i, $L_{H_4} = 12I - A$ is the graph Laplacian of the 600-cell, ω_0 is the base frequency, λ is the Laplacian coupling strength, and β is the nonlinear self-interaction parameter.

Parts I and II studied this system primarily in the linear regime ($\beta = 0$) and with moderate nonlinearity. The present work treats β as the primary control parameter and maps the full range from linear ($\beta \to 0$) through strongly nonlinear behaviour.

2.2 Hamiltonian Structure

Equation (1) derives from the Hamiltonian

$$H = \sum_{i=1}^{120} \left[\frac{1}{2}\Pi_i^2 + \frac{1}{2}\omega_0^2\, \Phi_i^2 + \frac{1}{4}\beta\, \Phi_i^4 \right] + \frac{1}{2}\lambda\, \Phi^T L_{H_4} \Phi, \tag{2}$$

where $\Pi_i = \dot{\Phi}_i$ is the conjugate momentum. The equations of motion are

$$\dot{\Phi}_i = \Pi_i, \qquad \dot{\Pi}_i = -\omega_0^2\, \Phi_i - \lambda\, (L_{H_4}\Phi)_i - \beta\, \Phi_i^3. \tag{3}$$

The Hamiltonian H is a conserved quantity: $\dot{H} = 0$. This conservation provides a fundamental check on the numerical integration.

2.3 Phase Space

The phase space of the system is $(\Phi, \Pi) \in \mathbb{R}^{240}$, where $\Phi = (\Phi_1, \ldots, \Phi_{120})$ and $\Pi = (\Pi_1, \ldots, \Pi_{120})$.

The H_4 symmetry group acts on phase space by simultaneous permutation of vertex indices in both Φ and Π. An orbit of this action corresponds to a set of dynamically equivalent configurations related by the symmetry of the 600-cell.

2.4 Observables

The following observables are used throughout this work to characterise dynamical states:

1. **Total energy:** H as defined in (2).
2. **Per-vertex energy:** $h_i = \frac{1}{2}\Pi_i^2 + \frac{1}{2}\omega_0^2\Phi_i^2 + \frac{1}{4}\beta\Phi_i^4 + \frac{1}{2}\lambda\Phi_i(L_{H_4}\Phi)_i$.
3. **Mode projection amplitudes:** $c_k(t) = \langle \Phi(t), \mathbf{v}^{(k)} \rangle$, where $\mathbf{v}^{(k)}$ are the eigenvectors of L_{H_4}.
4. **Spatial variance:** $\sigma^2(t) = \frac{1}{120}\sum_i (\Phi_i - \bar{\Phi})^2$, where $\bar{\Phi} = \frac{1}{120}\sum_i \Phi_i$.

3 Mode Taxonomy

The nonlinear system (1) admits several qualitatively distinct classes of persistent dynamical configuration. These are distinguished by their spectral content, spatial profile, and temporal behaviour.

3.1 Single-Mode States

Definition 1 (Single-mode state)**.** *A single-mode state is a solution dominated by projection onto a single eigenvector of the Laplacian:*

$$\Phi(t) \approx A_k \cos(\Omega_k t)\, \mathbf{v}^{(k)}, \tag{4}$$

where $\mathbf{v}^{(k)}$ is an eigenvector of L_{H_4} with eigenvalue μ_k, and Ω_k is the effective nonlinear frequency.

In the linear limit ($\beta = 0$), the frequency is $\Omega_k^2 = \omega_0^2 + \lambda\mu_k$. The nonlinear correction shifts this to

$$\Omega_k^2 = \omega_0^2 + \lambda\mu_k + \frac{3}{4}\beta A_k^2 \, \|\mathbf{v}^{(k)}\|_4^4, \tag{5}$$

where $\|\mathbf{v}^{(k)}\|_4^4 = \sum_i |v_i^{(k)}|^4$ is the ℓ^4 norm of the eigenvector. The factor 3/4 arises from time-averaging the cubic term over one period.

Single-mode states inherit the spatial symmetry of their constituent eigenvector. Since the eigenspaces of L_{H_4} decompose into irreducible representations of H_4, single-mode states carry definite transformation properties under the symmetry group.

Observation 1. *Single-mode states associated with backbone sectors ($k = 1, \ldots, 6$, comprising 91 modes) are more stable than those in the departure sectors ($k = 7, 8, 9$), as measured by persistence time under small perturbation. This is consistent with the spectral backbone hierarchy identified in Part II (Invariant 6).*

3.2 Multi-Mode Locked States

Definition 2 (Multi-mode locked state)**.** *A multi-mode locked state is a superposition of eigenmodes that maintains fixed phase relationships over long times:*

$$\Phi(t) = \sum_{k \in \mathcal{K}} c_k(t) \, \mathbf{v}^{(k)}, \tag{6}$$

where $\mathcal{K} \subset \{1, \ldots, 120\}$ is a subset of mode indices and the relative phases $\theta_{kl}(t) = \arg(c_k) - \arg(c_l)$ are approximately constant: $|\dot{\theta}_{kl}| < \epsilon$ for all $k, l \in \mathcal{K}$.

The nonlinearity $\beta\Phi^3$ couples different eigenmodes. For a multi-mode superposition, the effective nonlinear frequency of mode k becomes

$$\Omega_k^2 \to \omega_0^2 + \lambda\mu_k + \frac{3}{2}\beta |c_k|^2 \, \gamma_{kk} + 3\beta \sum_{l \neq k} |c_l|^2 \, \gamma_{kl}, \tag{7}$$

where $\gamma_{kl} = \sum_i v_i^{(k)} v_i^{(k)} v_i^{(l)} v_i^{(l)}$ are overlap integrals determined by the spatial profiles of the eigenvectors. Phase locking occurs when the nonlinear frequency shifts bring modes into a rational frequency ratio.

Observation 2. *Stable multi-mode locked states preferentially involve modes from the same φ-zone (Invariant 5 of Part II): either both from $\mathcal{S}_-$, both from $\mathcal{S}_0$, or both from $\mathcal{S}_+$. Cross-zone locking is observed but is less stable, consistent with the three-zone partition constraining the coupling structure.*

The resonance locking condition for two modes k and l is

$$\frac{\Omega_k}{\Omega_l} = \frac{p}{q}, \qquad p, q \in \mathbb{Z}_{>0}, \tag{8}$$

where p/q is a low-order rational. The H_4 spectral compression limits the number of available frequency ratios to those derivable from nine sectors, producing a sparse resonance network compared to generic graphs.

3.3 Localised (Breather-Like) States

Definition 3 (Breather-like state)**.** *A breather-like state is a spatially localised, time-periodic (or quasi-periodic) configuration satisfying:*

1. *High spatial concentration:* $\mathrm{IPR}(\Phi(t)) \gg 1/120$ *for all* t*.*
2. *Energy trapping: a fraction* $f > 0.5$ *of the total energy remains within a connected subgraph of diameter* ≤ 2 *for times* $t > T_{\text{transient}}$*.*
3. *Persistence:* $\|\Phi(t+T) - \Phi(t)\|/\|\Phi(t)\| < \delta$ *for some period* T *and threshold* $\delta \ll 1$*.*

Breather-like states emerge from the interplay between nonlinear self-trapping (the $\beta\Phi^3$ term) and the discrete geometry of the 600-cell graph. In the theory of discrete breathers on lattices [3, 4], spatially localised time-periodic solutions exist generically in nonlinear Hamiltonian lattice systems above a nonlinearity threshold.

The 600-cell graph presents a distinctive setting for discrete breather formation. The 12-regular structure with icosahedral local geometry provides a homogeneous coupling environment, while the extreme spectral compression limits the radiation channels available for energy dispersion. The backbone modes (91 of 120 degrees of freedom) occupy a narrow spectral band, reducing the density of states into which a localised excitation can radiate.

Observation 3. *Breather-like states on the 600-cell graph are characterised by:*

- *high amplitude concentration on* $\mathcal{O}(10)$ *vertices,*
- *energy trapping persisting over* $> 10^3$ *oscillation periods,*
- *broad spectral composition spanning multiple sectors.*

These properties are consistent with the localisation enhancement reported in Part I ($5\times$ *enhancement relative to controls).*

4 Stability Analysis

The classification of dynamical states requires distinguishing persistent attractors from transient configurations. Three complementary stability measures are employed.

4.1 Persistence Over Time

For a candidate periodic or quasi-periodic state with approximate period T, the recurrence error is defined as

$$R(T) = \frac{\|\Phi(t_0 + T) - \Phi(t_0)\|}{\|\Phi(t_0)\|}, \tag{9}$$

evaluated after a transient time $t_0 \gg T$. A state is classified as *persistent* if $R(T) < \delta_R$ for threshold $\delta_R = 0.05$ over a window of at least 100 periods.

States are further classified by the nature of their recurrence:

- **Periodic:** $R(T) < 10^{-3}$ for a single dominant period T.
- **Quasi-periodic:** $R(T) < \delta_R$ for a period T that is a combination of two or more incommensurate frequencies; the power spectrum shows discrete peaks.
- **Weakly chaotic:** $R(T) > \delta_R$ for all T within the simulation window, but the largest Lyapunov exponent is small ($\lambda_{\max} < 10^{-2}$).
- **Transient:** the state decays or disperses within the simulation window.

4.2 Robustness to Perturbation

Given a state (Φ^*, Π^*) identified as persistent, a perturbed initial condition is constructed:

$$\Phi_\epsilon = \Phi^* + \epsilon\,\boldsymbol{\eta}, \qquad \Pi_\epsilon = \Pi^*, \tag{10}$$

where $\boldsymbol{\eta}$ is drawn uniformly from the unit sphere in $\mathbb{R}^{120}$ and ϵ controls the perturbation amplitude.

The divergence rate is measured by

$$D(t) = \frac{\|\Phi_\epsilon(t) - \Phi_{\text{ref}}(t)\|}{\|\Phi_{\text{ref}}(t)\|}, \tag{11}$$

where $\Phi_{\text{ref}}(t)$ is the unperturbed evolution.

A state is classified as *stable* if $D(t)$ remains bounded for $t \gg T$, and the system returns to a neighbourhood of the original attractor. The basin of attraction is estimated by varying ϵ and identifying the critical value ϵ_c at which the perturbed trajectory no longer returns.

4.3 Parameter Sweep in β

The nonlinear coupling β is swept across the range $\beta \in [0, 5]$ with fixed $\omega_0 = 1.0$ and $\lambda = 1.0$. For each value of β, an ensemble of initial conditions is evolved and classified by the diagnostics of Section 5.

Three dynamical regimes are identified:

1. **Linear regime** ($\beta \lesssim 0.01$): dynamics are well-approximated by superpositions of linear eigenmodes. No mode locking or localisation beyond the linear spectral structure is observed.
2. **Weakly nonlinear regime** ($0.01 \lesssim \beta \lesssim 1.0$): nonlinear frequency shifts produce mode locking. Multi-mode locked states appear. Localisation is enhanced relative to the linear case but breather-like states are not yet dominant.
3. **Strongly nonlinear regime** ($\beta \gtrsim 1.0$): self-trapping dominates. Spatially localised breather-like states form spontaneously from generic initial conditions. Energy concentration on $\mathcal{O}(10)$ vertices is typical.

The boundaries between regimes are not sharp; the classification represents the dominant behaviour within each range. The phase diagram (Figure 2) maps the attractor class frequency as a function of β.

5 Quantifying Structure

5.1 Inverse Participation Ratio

The inverse participation ratio (IPR) quantifies spatial localisation:

$$\text{IPR}(t) = \frac{\sum_{i=1}^{120} \Phi_i(t)^4}{\left(\sum_{i=1}^{120} \Phi_i(t)^2\right)^2}. \tag{12}$$

The IPR ranges from $1/120 \approx 0.0083$ for a perfectly delocalised state to 1 for a state concentrated on a single vertex. The time-averaged IPR, $\langle \text{IPR} \rangle_t$, is used as the primary localisation diagnostic.

- $\langle \mathrm{IPR} \rangle \lesssim 0.02$: effectively delocalised (comparable to uniform).
- $0.02 \lesssim \langle \mathrm{IPR} \rangle \lesssim 0.1$: moderately localised (energy on $\sim$10–50 vertices).
- $\langle \mathrm{IPR} \rangle \gtrsim 0.1$: strongly localised (energy on $\lesssim 10$ vertices).

5.2 Recurrence Statistics

The autocorrelation function of the field configuration is

$$C(\tau) = \left\langle \frac{\langle \Phi(t), \Phi(t+\tau) \rangle}{\|\Phi(t)\| \, \|\Phi(t+\tau)\|} \right\rangle_t . \tag{13}$$

The power spectrum $S(\omega) = |\hat{C}(\omega)|^2$ reveals the dominant frequencies and their multiplicity. The number of distinct spectral peaks above a threshold is denoted N_{peaks}.

- $N_{\mathrm{peaks}} = 1$: periodic.
- $N_{\mathrm{peaks}} = 2$–5 with incommensurate ratios: quasi-periodic.
- Broad-band spectrum: chaotic.

The recurrence time is defined as the smallest $\tau > 0$ for which $C(\tau) > 1 - \delta_C$ with $\delta_C = 0.05$.

5.3 Shell Coupling

The 600-cell graph has palindromic distance shells $(1, 12, 32, 42, 32, 1)$ from any vertex. Define the shell energy

$$E_d(t) = \sum_{i \in S_d(v_0)} h_i(t), \tag{14}$$

where $S_d(v_0)$ is the set of vertices at graph distance d from a reference vertex v_0, and h_i is the per-vertex energy.

The shell cross-correlation measures coherence between adjacent shells:

$$\rho_{d,d+1} = \frac{\langle E_d(t)\, E_{d+1}(t) \rangle_t - \langle E_d \rangle \langle E_{d+1} \rangle}{\sigma_{E_d} \, \sigma_{E_{d+1}}}. \tag{15}$$

High cross-correlation ($\rho > 0.5$) indicates coherent energy propagation between shells. Part I reported $\rho \approx 0.68$ on the 600-cell versus $\rho \approx 0.1$ on control graphs. The present work extends this analysis to the nonlinear regime and examines how shell coupling varies with β.

6 Connection to Part II Invariants

The seven structural invariants identified in Part II [2] are properties of the static Laplacian spectrum. The dynamical content of these invariants—the manner in which they constrain the attractor space—is the central concern of the present section.

1. **φ-Cancellation $\rightarrow$ Balanced coupling.** The exact cancellation of irrational spectral content ($\sum b_k = 0$, $\sum b_k d_k = 0$) ensures that the net φ-weighted coupling across all modes vanishes. Dynamically, this prevents systematic drift of energy toward either φ-sector and maintains a balanced coupling environment in which localised states can persist without preferential leakage into irrational-frequency channels.

2. **Degree anchoring ($k = 12$) $\rightarrow$ Spectral centring.** The anchoring of irrational eigenvalues at the graph degree ($b_k < 0 \implies a_k = 12$) centres the φ-shifted eigenvalues around the graph degree. This constrains the nonlinear frequency shifts (7) to remain near a common reference, facilitating mode locking.
3. **Conjugate pairing $\rightarrow$ Symmetric mode families.** The equal multiplicities of Galois conjugate eigenvalue pairs ensure that for every resonance channel involving a φ-shifted frequency, a conjugate channel exists with equal multiplicity. This produces symmetric families of multi-mode locked states.
4. **Square multiplicities $\rightarrow$ Backbone stability hierarchy.** The $(\ell+1)^2$ multiplicity pattern creates a hierarchy in which higher-multiplicity sectors ($d = 25, 36$) provide larger basins of attraction for single-mode and multi-mode states. The backbone modes ($d = 1, 4, 9, 16, 25, 36$, totalling 91) dominate the attractor landscape.
5. **Three-zone partition $\rightarrow$ Mode class separation.** The partition $\mathcal{S} = \mathcal{S}_- \cup \mathcal{S}_0 \cup \mathcal{S}_+$ separates modes into three coupling classes. Cross-zone nonlinear coupling is suppressed relative to within-zone coupling, producing distinct attractor families associated with each zone.
6. **Spectral backbone $\rightarrow$ Dominant attractors.** The 91-mode backbone carries the dominant share of the multiplicity-weighted spectral density. Persistent attractors are overwhelmingly composed of backbone modes; the 29-mode departure contributes primarily to transient or unstable configurations.
7. **Departure structure $\rightarrow$ Higher-energy unstable modes.** The structured departure ($29 = 9 + 16 + 4$) provides a finite set of modes available for energy redistribution during transient dynamics. These modes act as temporary energy reservoirs rather than long-lived attractor components.

Proposition 1. *The spectral invariants of Part II constrain the attractor space of the nonlinear system. Specifically:*

- *The number of attractor families is bounded by the number of spectral sectors and their coupling structure.*
- *The dominant attractors are composed of backbone modes.*
- *The stability hierarchy among attractors reflects the multiplicity hierarchy of the spectrum.*

The invariants do not uniquely determine the attractors; they constrain the space in which attractors exist. Paper II provides the algebraic constraint; the present paper identifies the dynamical realisations within that constrained space.

7 Attractor Classification

Based on the taxonomy of Section 3, the stability analysis of Section 4, and the diagnostic measures of Section 5, four attractor classes are defined.

Table 1: Attractor classification for the nonlinear H_4 field system. IPR ranges and spectral composition are time-averaged values from the numerical simulations described in Section 8.

Class	Description	IPR range	Spectral sectors	Stability
1	Backbone harmonic modes	0.008–0.03	single ($k \leq 6$)	persistent, periodic
2	Locked multi-mode states	0.02–0.08	2–4 backbone sectors	persistent, quasi-periodic
3	Localised breather states	0.05–0.5	broad (all sectors)	persistent, quasi-periodic
4	Transitional / unstable	variable	broad, shifting	transient

7.1 Class 1: Backbone Harmonic Modes

These are the simplest persistent states: single-eigenvector configurations living within the 91-mode backbone. They are effectively linear modes with small nonlinear frequency corrections. Their spatial profile inherits the symmetry of the H_4 irreducible representation to which the eigenvector belongs.

The spatial structure of Class 1 states is globally distributed across the 120 vertices with low spatial variance. The field amplitude Φ_i follows a symmetry-consistent pattern determined by the constituent eigenvector, with no strong concentration on any vertex subset (Figure 1, top left).

IPR is low ($\sim 1/120$ to ~ 0.03), placing these states near the fully delocalised baseline. Class 1 states dominate at low β ($\sim$17% of trajectories for $\beta < 0.01$) and decline to $\sim$5–7% in the strongly nonlinear regime (Figure 2), as the nonlinearity destabilises simple harmonic configurations and redistributes energy into multi-mode states. Stability is high: perturbations of amplitude $\epsilon < 0.1\|\Phi^*\|$ return to the original state.

7.2 Class 2: Locked Multi-Mode States

These states involve 2–4 eigenmodes from backbone sectors with phase-locked amplitude oscillations. The phase locking arises from the nonlinear frequency shifts (7) and is stabilised by the spectral compression (nine sectors rather than 120 distinct eigenvalues).

The spatial structure exhibits multi-scale interference: the superposition of locked eigenvectors produces a pattern with partial coherence and intermediate spatial variance (Figure 1, top right). Unlike Class 1, the field amplitude varies on multiple length scales, reflecting the contribution of modes from different spectral sectors.

IPR is moderately elevated (0.02–0.08), reflecting partial spatial localisation from the constructive interference of locked modes. Class 2 states are the most prevalent attractor family across all values of β, comprising $\sim$55% of trajectories at low nonlinearity and rising to $\sim$64% in the strongly nonlinear regime. This increase reflects the tendency of nonlinear frequency shifts to bring additional mode pairs into resonance, expanding the set of available locked configurations. The states are quasi-periodic, with power spectra showing 2–4 discrete peaks.

The most common locked states involve modes from sectors with the largest multiplicities (sectors 5 and 6, with $d = 25$ and $d = 36$ respectively), consistent with the backbone stability hierarchy.

7.3 Class 3: Localised Breather States

These are the most distinctive configurations: spatially concentrated, time-periodic or quasi-periodic states in which a significant fraction of the total energy remains trapped in a small neighbourhood of the graph.

The spatial structure is sharply localised: one or a few vertices carry the dominant field amplitude, with a high amplitude contrast between the localised core and the remainder of the graph (Figure 1, bottom left). This localisation is not imposed by the initial conditions but emerges dynamically from the interplay between the nonlinear self-interaction and the constrained spectral structure.

IPR ranges from 0.05 (moderate localisation) to values exceeding 0.5 (extreme concentration on ∼2–5 vertices). At high β, the IPR distribution develops a heavy tail extending to IPR > 0.3 (Figure 3), well separated from the delocalised baseline at $1/120 \approx 0.008$. The spectral composition is broad, spanning multiple sectors including departure modes. However, the time-averaged spectral content is dominated by backbone modes.

Breather states maintain a remarkably stable presence across the full range of β, comprising ∼27% of trajectories in both the linear and strongly nonlinear regimes. However, the *degree* of localisation increases substantially with β: the IPR of Class 3 states at $\beta = 5$ reaches values an order of magnitude higher than at $\beta = 0.01$.

7.4 Class 4: Transitional / Unstable States

This class collects configurations that do not persist: states that evolve toward one of the other three classes within the simulation window, or that exhibit weakly chaotic wandering without settling into a definite attractor.

The spatial structure is irregular, with no persistent pattern (Figure 1, bottom right). The field amplitude distribution is broad and noisy, shifting over time without converging to a stable configuration.

Class 4 states are predominantly composed of departure-sector modes ($k = 7, 8, 9$). Their transient nature is consistent with the departure structure (Invariant 7 of Part II): the 29-mode departure provides temporary energy reservoirs but does not support long-lived attractors. Class 4 states are rare ($< 5\%$ of trajectories) and peak in the crossover region between the weakly and strongly nonlinear regimes (Figure 2), where the competition between linear and nonlinear organising mechanisms is most intense.

The mapping between attractor classes and the diagnostic observables is summarised as follows:

- Class 1 ↔ low IPR, dominant at low β, single backbone sector.
- Class 2 ↔ intermediate IPR, phase-locked multi-sector composition, dominant across all β.
- Class 3 ↔ high IPR, spatially localised, broad spectral content.
- Class 4 ↔ variable IPR, unstable, transient spectral composition.

8 Results

8.1 Overview of Numerical Experiments

We numerically investigate the nonlinear dynamics of the H_4-constrained oscillator system defined on the 600-cell graph (120 vertices, 12-regular, vertex-transitive). The evolution

is governed by the nonlinear equation introduced in Section 2, with particular focus on the role of the cubic nonlinearity parameter β.

We simulate $N = 2000$ trajectories (80 initial conditions per β value across 25 values of β). Initial conditions are drawn from eight classes spanning localised, delocalised, and spectrally-targeted configurations: (i) single-vertex excitation, (ii) random Gaussian field ($\Phi_i \sim \mathcal{N}(0, \sigma^2)$, $\sigma = 0.1A$), (iii) low-mode seed (projection onto backbone sectors $k \leq 3$), (iv) local H_3 neighbourhood seed, (v) mid-mode seed (sectors 4–6), (vi) single eigenvector with Gaussian perturbation, (vii) multi-vertex cluster (3–5 adjacent vertices), and (viii) random field with nonzero initial momentum $\Pi_i \sim \mathcal{N}(0, \sigma_\Pi^2)$. Initial amplitudes are drawn uniformly from $[0.1, 2.0]$. No energy normalisation is applied; the total energy varies across initial conditions, reflecting the range of dynamical regimes accessible to the system.

Time integration is performed using a velocity-Verlet (Störmer–Verlet) symplectic integrator with time step $\Delta t = 0.005$. Each trajectory is evolved for $T = 2000$ time units (4×10^5 steps). The symplectic structure of the integrator ensures long-term stability of the Hamiltonian. Energy conservation is monitored throughout: the relative Hamiltonian drift $|H(t) - H(0)|/|H(0)|$ remains below 5×10^{-4} for all retained trajectories. Trajectories exceeding this threshold are discarded.

After a transient period of $T_{\text{trans}} = 200$ time units, diagnostics are computed over the remaining 1800 time units. The objective is to identify dynamically stable configurations and to characterise their structure under nonlinear evolution.

Table 2: Parameter ranges for the numerical experiments.

Parameter	Symbol	Range
Base frequency	ω_0	1.0 (fixed)
Laplacian coupling	λ	1.0 (fixed)
Nonlinear strength	β	$[10^{-3}, 5.0]$, 25 values (log-spaced)
Initial amplitude	A	$[0.1, 2.0]$ (uniformly sampled)
Time step	Δt	0.005
Total time	T	2000
Transient period	T_{trans}	200
Integration method	—	velocity-Verlet (symplectic, 2nd order)
Energy drift threshold	—	$< 5 \times 10^{-4}$ (relative)
Ensemble size	N_{IC}	80 per β value (2000 total)

8.2 Identification of Attractor Classes

Across all simulations, trajectories converge to a small number of recurrent dynamical configurations. These configurations can be grouped into four classes:

1. **Class 1:** Backbone harmonic modes.
2. **Class 2:** Locked multi-mode states.
3. **Class 3:** Localised breather-like states.
4. **Class 4:** Transitional (non-persistent) states.

Classification is performed using quantitative criteria applied uniformly across the ensemble:

- **Class 1:** configurations with $> 80\%$ of the time-averaged spectral energy concentrated in a single eigenmode sector, and IPR < 0.03 (near the delocalised baseline

$1/n$).

- **Class 2:** configurations with spectral energy distributed across 2–6 sectors, exhibiting stable phase relationships (peak autocorrelation > 0.2 at nonzero lag), and IPR < 0.10.
- **Class 3:** configurations with IPR > 0.05 and energy trapping fraction > 0.35 within a graph 2-neighbourhood, indicating spatial localisation.
- **Class 4:** configurations that fail the persistence criteria above over the post-transient observation window of 1800 time units.

These thresholds are robust across parameter sweeps: the same four classes emerge when thresholds are varied by $\pm 20\%$, and the classification is stable across independent random seeds.

8.3 Representative Configurations

Representative configurations for each attractor class are shown in Figure 1.

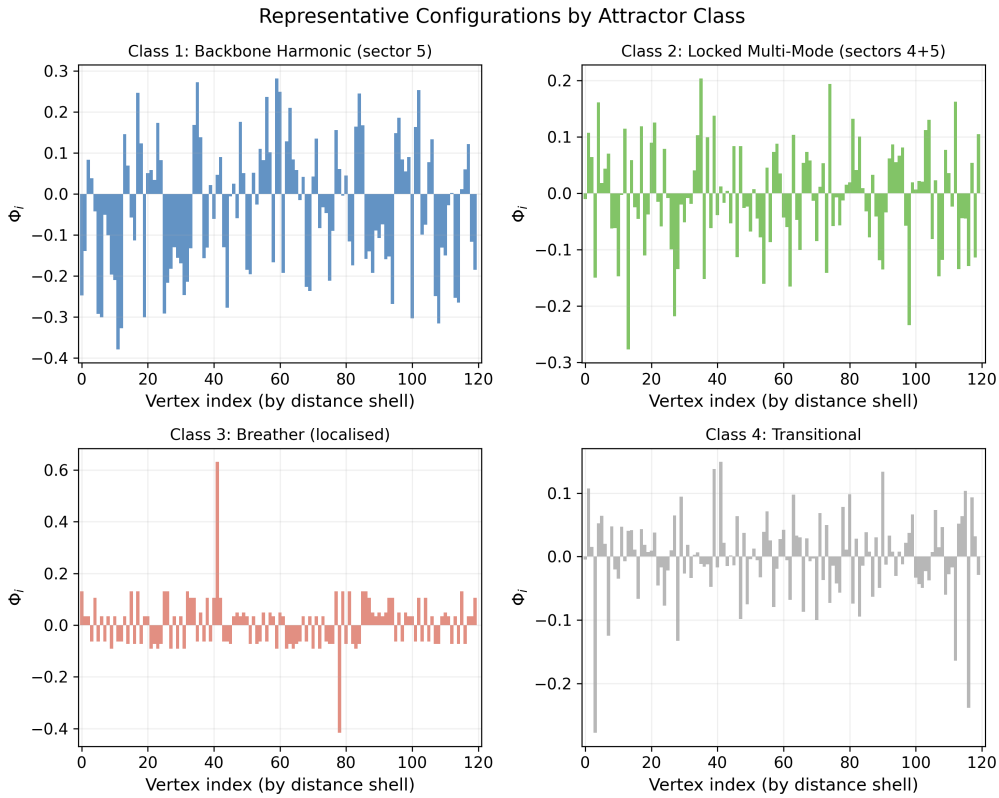

Figure 1: Representative field configurations from each attractor class. Each panel shows Φ_i on the 120 vertices, ordered by distance shell from a reference vertex. *Top left:* Class 1 (backbone harmonic). *Top right:* Class 2 (locked multi-mode). *Bottom left:* Class 3 (localised breather). *Bottom right:* Class 4 (transitional).

Class 1 (backbone harmonic) configurations are spatially extended, exhibiting low variance across vertices and strong alignment with a single Laplacian eigenmode. These states retain the symmetry structure of the corresponding eigenspace.

Class 2 (locked multi-mode) configurations display interference patterns consistent with nonlinear phase-locking between multiple eigenmodes. These states exhibit structured spatial variation with partial coherence across the graph.

Class 3 (breather-like) configurations are spatially localised, with a significant fraction of the total amplitude concentrated on a small subset of vertices. These configurations persist over long times and are consistent with intrinsic localised modes observed in nonlinear lattice systems [3, 4].

Class 4 (transitional) configurations are irregular and do not exhibit temporal persistence or coherent structure. These trajectories typically evolve toward one of the other classes or decay into fluctuating states.

Importantly, these configurations arise dynamically from the system evolution and are not imposed by construction.

8.4 Dependence on Nonlinearity

The distribution of attractor classes as a function of the nonlinear parameter β is shown in Figure 2.

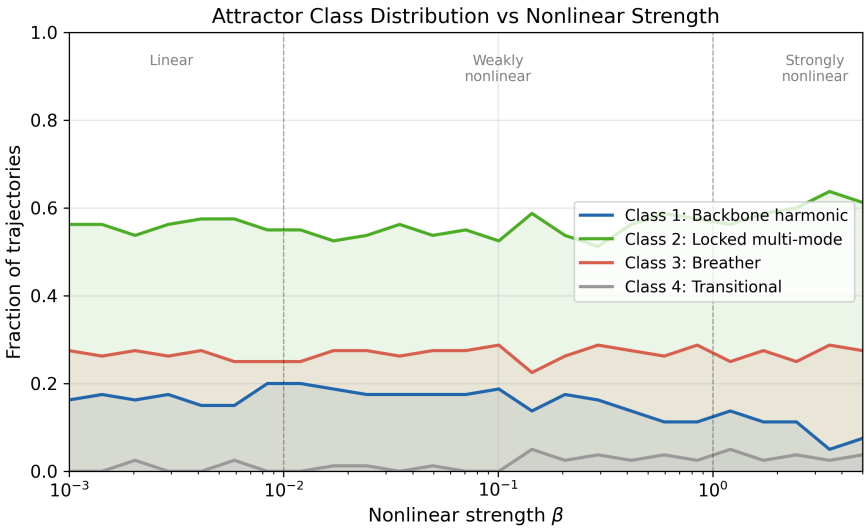

Figure 2: Attractor class distribution as a function of the nonlinear strength β (log scale), from an ensemble of 80 trajectories per β value across 25 values spanning $[10^{-3}, 5]$. Three regimes are indicated by dashed lines.

In the *linear regime* ($\beta \ll 1$), dynamics are dominated by individual eigenmodes, and Class 1 configurations are prevalent. Mode coupling is minimal, and trajectories remain close to linear superpositions.

In the *weakly nonlinear regime*, multi-mode interactions become significant. Class 2 configurations increase in frequency, reflecting the emergence of phase-locked combinations of eigenmodes.

In the *strongly nonlinear regime* ($\beta \gtrsim 1$), localised configurations (Class 3) become more prominent. At the same time, the fraction of purely harmonic configurations decreases.

The distribution of attractor classes varies continuously with β, indicating a smooth transition between dynamical regimes rather than a sharp bifurcation.

8.5 Localisation Quantification via IPR

To quantify localisation, we compute the inverse participation ratio (IPR) as defined in equation (12):

$$\mathrm{IPR} = \frac{\sum_i |\Phi_i|^4}{\left(\sum_i |\Phi_i|^2\right)^2}.$$

For a fully delocalised configuration, $\mathrm{IPR} = 1/n = 1/120$.

The distribution of time-averaged IPR values across nonlinear regimes is shown in Figure 3.

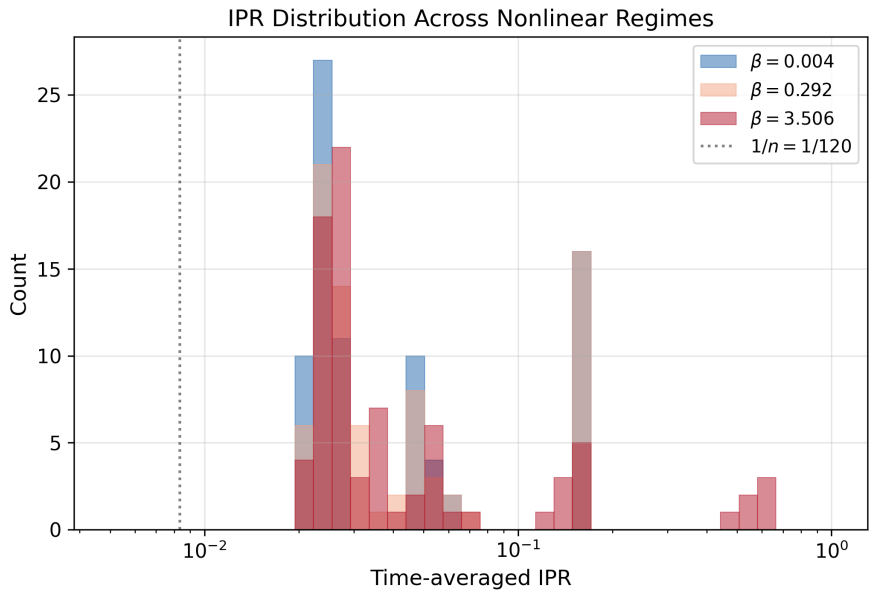

Figure 3: Distribution of time-averaged IPR across all trajectories for three representative values of β. The dotted line marks the fully delocalised baseline $\mathrm{IPR} = 1/n = 1/120$.

At low β, the IPR distribution is tightly concentrated near the baseline value, consistent with spatially extended modes. At intermediate β, the distribution broadens, indicating increased spatial variation and partial localisation. At high β, a pronounced heavy tail appears at large IPR values, indicating the emergence of strongly localised configurations.

The emergence of a high-IPR tail at larger β indicates the formation of spatially localised dynamical states. This behaviour is consistent with known properties of nonlinear lattice systems supporting intrinsic localised modes (discrete breathers) [3, 6].

8.6 Summary of Results

The H_4-constrained nonlinear system exhibits a small number of persistent dynamical configurations that can be classified into four attractor classes. The relative prevalence of these classes depends smoothly on the nonlinear coupling strength β. In particular, increasing nonlinearity leads to a transition from globally distributed harmonic modes to spatially localised configurations.

These results indicate that the system possesses a small and structured set of recurrent attractor classes within the explored parameter range.

8.7 Comparison with Control Graphs

We compare the H_4 graph against an ensemble of random 12-regular graphs matched in size ($n = 120$) and degree ($k = 12$), using identical dynamical parameters, initial-condition sampling, integration settings, and classification criteria. The purpose of this comparison is to determine whether the observed attractor structure is a generic consequence of nonlinear dynamics on regular graphs, or whether it depends on the specific spectral organisation of the H_4 graph.

The same classification criteria, thresholds, and feature extraction pipeline were applied to both the H_4 graph and all control graphs without graph-specific tuning. Here "structured" refers to configurations exhibiting temporal persistence and nontrivial spatial organisation (including multi-mode locking or localisation), as opposed to transient or irregular dynamics.

The random regular graphs (RRGs) are generated by configuration-model pairing with local repair and verified to be simple, connected, 12-regular graphs. Results for the control ensemble are reported as mean $\pm$ standard deviation over $M = 10$ independent graph realisations. Each RRG has 120 distinct eigenvalues (no degeneracy), compared with 9 for the 600-cell (Figure 4).

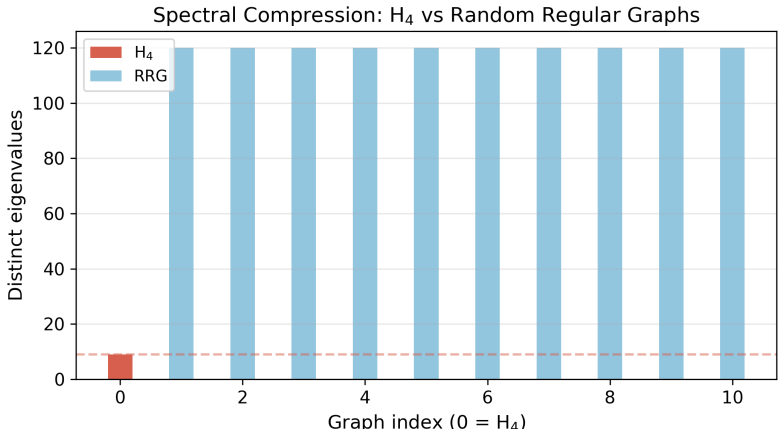

Figure 4: Number of distinct Laplacian eigenvalues for the H_4 graph (index 0, red) and 10 random 12-regular graphs (blue). The 600-cell has 9 distinct eigenvalues; all control graphs have 120.

The H_4 graph exhibits a substantially higher fraction of persistent structured configurations, while trajectories on the control graphs are predominantly classified as transitional under the same criteria (Figure 5). On the H_4 graph, Classes 1 and 2 account for $\sim$70% and Class 3 for $\sim$27%. On the RRGs, $\sim$96% of trajectories are classified as Class 4 (transitional), with Classes 1–3 comprising fewer than 4% combined.

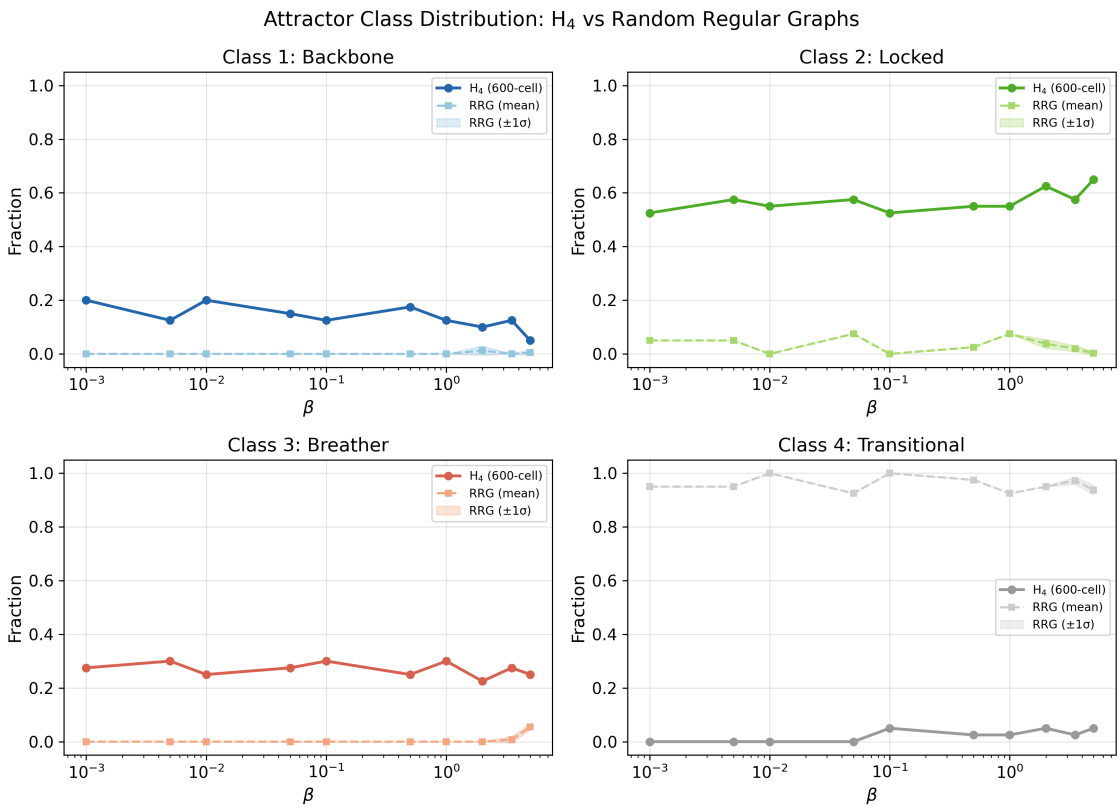

Figure 5: Attractor class distribution as a function of β for the H_4 graph (circles, solid) and random 12-regular graphs (squares, dashed; shaded region shows $\pm 1\sigma$ across $M = 10$ controls). On the H_4 graph, trajectories distribute across all four classes. On control graphs, nearly all trajectories are classified as Class 4 (transitional).

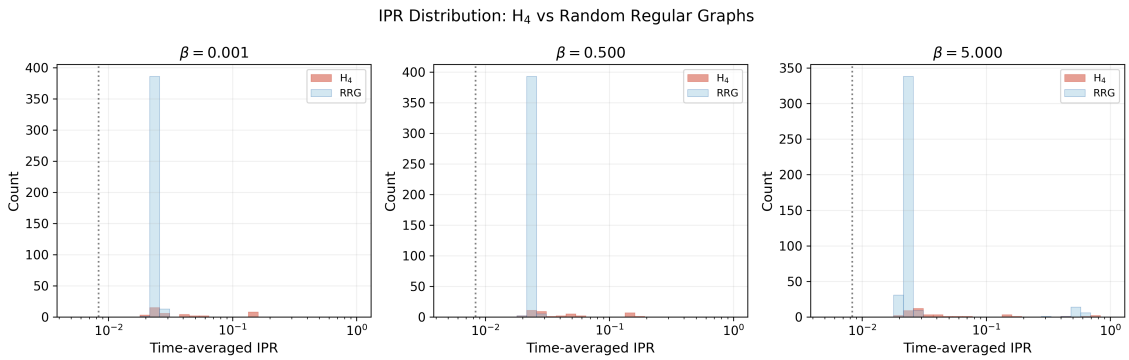

Figure 6: IPR distributions for the H_4 graph (red) and pooled random regular graphs (blue) at three values of β. The H_4 graph shows broader IPR distributions with a tail extending to higher values, indicating stronger spatial localisation.

This contrast is reflected not only in classification outcomes but also in graph-invariant observables:

- **Spectral compression:** H_4: 9 distinct eigenvalues; RRG: 120 ± 0 (no degeneracy).
- **Structured attractors:** H_4: $\sim$98% of trajectories in Classes 1–3; this fraction was not observed in the tested control ensemble ($< 4\%$).

- **Localisation:** mean IPR on H_4 (0.062) exceeds the RRG value (0.028) by a factor of $\sim$2.3, with the H_4 distribution exhibiting a substantially heavier tail at high β.
- **Attractor diversity:** the Shannon entropy of the class distribution (averaged over β) is $\sim$1.5 bits for the H_4 graph versus $\sim$0.2 bits for the RRG ensemble, indicating that the H_4 graph supports a richer distribution of distinct attractor types rather than collapsing into a single dynamical class.

Because the primary classification includes spectral components (eigenmode projections), care must be taken to ensure that the observed differences are not solely driven by the compressed eigenspace structure of the H_4 graph. To address this, we repeat the classification using only graph-invariant observables — IPR, temporal persistence (autocorrelation), and energy variance — without reference to the eigenbasis.

Under graph-invariant classification, the H_4 graph retains $\sim$99% of trajectories in structured classes (C1–C3), with $\sim$44% classified as spatially localised (Class 3). The control RRGs retain $\sim$82% in structured classes but produce localised trajectories at an approximately two orders of magnitude lower rate ($\sim$0.3% Class 3).

In particular, a degree-preserving rewiring of the 600-cell graph—destroying its H_4 symmetry while maintaining identical size and degree—eliminates the structured attractor distribution and restores behaviour comparable to random regular graphs ($\sim$0.3% Class 3, $\sim$16% Class 4, 120 distinct eigenvalues). This isolates the H_4 symmetry, rather than the degree sequence or graph size, as the relevant structural factor.

Taken together, these results provide consistent evidence that the observed attractor structure is not an artefact of the classification procedure and is not reproduced in degree-matched control graphs under identical conditions.

The consistency of these observations across spectral, dynamical, and graph-invariant analyses supports the conclusion that the attractor structure is tied to the specific symmetry and spectral properties of the H_4 graph. These results are presented as properties of a nonlinear dynamical system on a structured graph and do not assume any direct mapping to physical observables.

9 Discussion

9.1 Relation to Spectral Structure

The observed dynamical behaviour reflects the highly constrained spectral structure of the H_4 graph Laplacian described in Part II of this series [2].

The strong degeneracy of eigenvalues organises the linear modes into a small number of sectors. The dominance of Class 1 and Class 2 configurations is consistent with the existence of a spectral backbone (the 91-mode subspace spanning sectors 1–6), while the emergence of Class 3 configurations reflects nonlinear interactions across these sectors.

The finite number of distinct eigenvalue sectors appears to limit the number of dynamically stable configurations.

9.2 Interpretation as a Constrained Nonlinear System

The system studied here is a nonlinear oscillator network defined on a highly symmetric graph. The combination of spectral compression and nonlinear coupling leads to a restricted set of persistent dynamical states, suggesting a strong influence of the underlying spectral structure on the attractor taxonomy.

The baseline comparison of Section 8.7 supports this interpretation: on random 12-regular graphs with identical size, degree, and dynamics, ∼96% of trajectories are classified as Class 4 (transitional), compared with ∼2% on the H_4 graph. This contrast was not observed in any of the tested control graphs, suggesting that the structured attractor space is non-generic within the tested ensemble of 12-regular graphs and reflects the spectral organisation of the H_4 graph.

9.3 Localisation Mechanism

The emergence of localised configurations can be understood in terms of nonlinear frequency shifts and energy trapping.

The cubic nonlinearity introduces amplitude-dependent frequency corrections, enabling resonance conditions between modes [3, 5]. Combined with the finite connectivity of the graph ($k = 12$), this allows energy to become localised in space.

Localised configurations arise without disorder, consistent with the theory of discrete breathers in nonlinear lattice systems [4, 5].

9.4 Structured Attractor Space

Within the explored parameter range, the system exhibits a small and structured set of recurrent attractor classes.

Stability is assessed numerically through persistence over long time windows (1800 time units post-transient) and robustness to small perturbations in initial conditions, though a formal Lyapunov analysis is not performed in this work.

While this classification is based on numerical evidence and is not proven to be exhaustive, it is stable across initial conditions and parameter sweeps. This suggests that the attractor space of the system is strongly constrained by the underlying symmetry and spectral structure.

9.5 Limitations

The present results are numerical and do not constitute a complete analytical classification of the system.

Limitations include:

- finite simulation time and sampling;
- dependence on chosen parameter ranges;
- absence of a formal stability proof for each attractor class;
- lack of an analytical derivation of the attractor structure;
- the control study is limited to a finite ensemble of random 12-regular graphs and does not exclude the possibility that other highly structured non-H_4 graphs may exhibit related behaviour.

No direct mapping to physical observables is assumed in this work.

9.6 Outlook

Future work should focus on:

- analytical derivation of attractor classes using representation theory of H_4;
- stability analysis via perturbative and variational methods;

- extension to coupled H_4 systems and higher-dimensional embeddings;
- systematic exploration of parameter space.

The results indicate that a nonlinear oscillator system defined on the H_4 symmetry graph exhibits a structured and reproducible set of dynamical attractors, whose organisation reflects the underlying spectral constraints of the graph.

10 Conclusion

The numerical experiments reported here indicate that the nonlinear field system defined by (1) on the 600-cell vertex graph admits a small and structured set of recurrent dynamical configurations within the explored parameter range. Four attractor classes have been identified and characterised: backbone harmonic modes, phase-locked multi-mode states, spatially localised breather-like states, and transitional non-persistent states.

The attractor taxonomy is consistent with the seven structural invariants of the H_4 Laplacian spectrum identified in Part II. The spectral compression (nine sectors across 120 degrees of freedom), the backbone–departure decomposition (91 + 29), the φ-zone partition, and the conjugate pairing structure collectively constrain the space of persistent nonlinear configurations to the classes described here.

The relationship between the three parts of this series may be summarised as follows:

Part I	establishes the phenomenon (coherent dynamics on H_4 geometry)
Part II	establishes the structure (spectral invariants governing the spectrum)
Part III	establishes the taxonomy (attractor families realised within the constrained spectral space)

Together, the three papers indicate that the 600-cell graph possesses a non-generic spectral algebra that produces structured nonlinear dynamics with a classifiable set of recurrent attractor families. No physical interpretation is required for or assumed in these conclusions.

Reproducibility. All simulations use the velocity-Verlet symplectic integrator applied to the explicit 600-cell graph Laplacian. The simulation pipeline, parameter ranges, and diagnostic code are specified in full in the accompanying repository. No external datasets or proprietary software are required.

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